



Deep Learning

Course code:

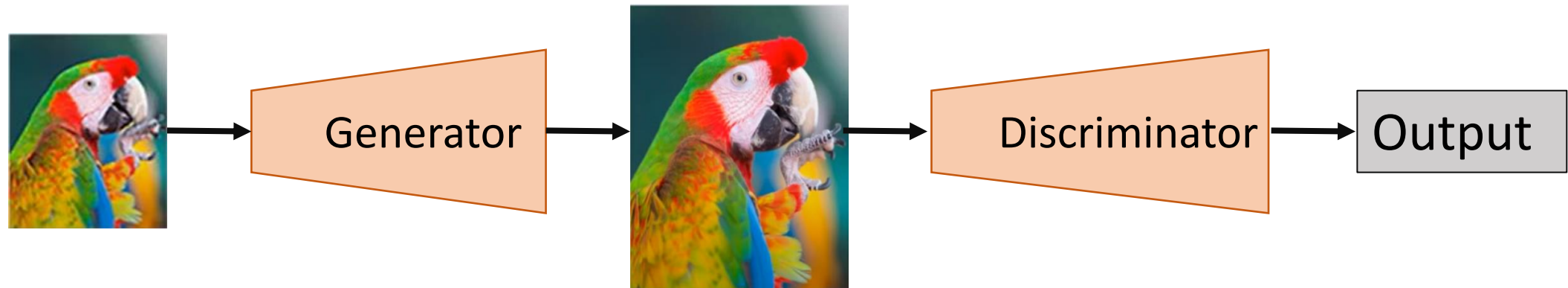
Unit 5: Autoencoders and
Generative Networks

Lecture 1: Image Data Generation
Using GAN Architectures

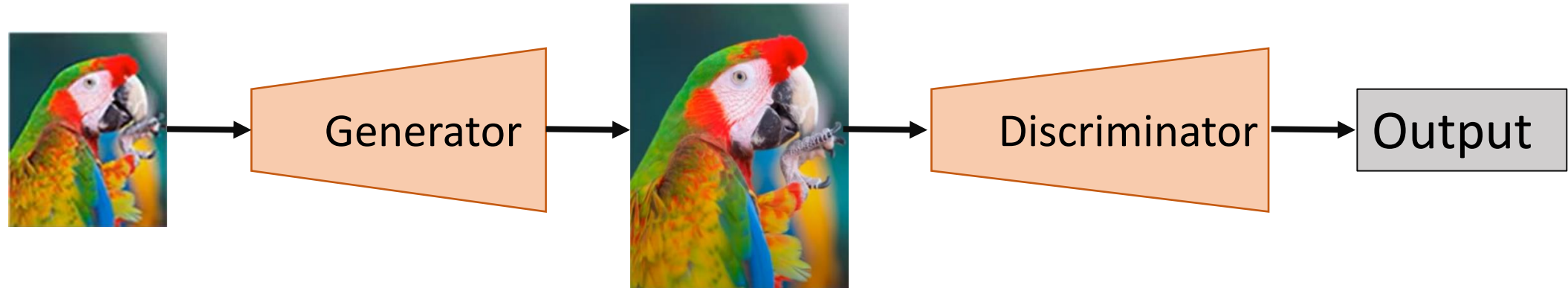




Super Resolution GAN

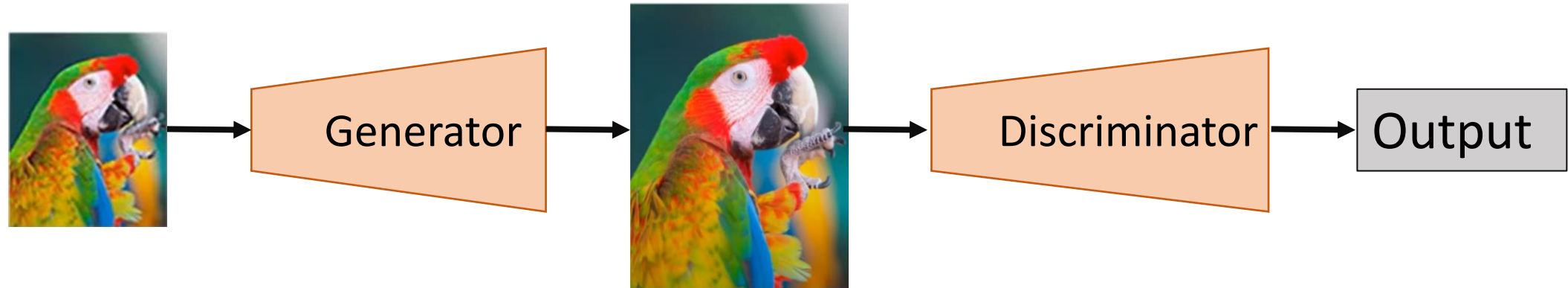


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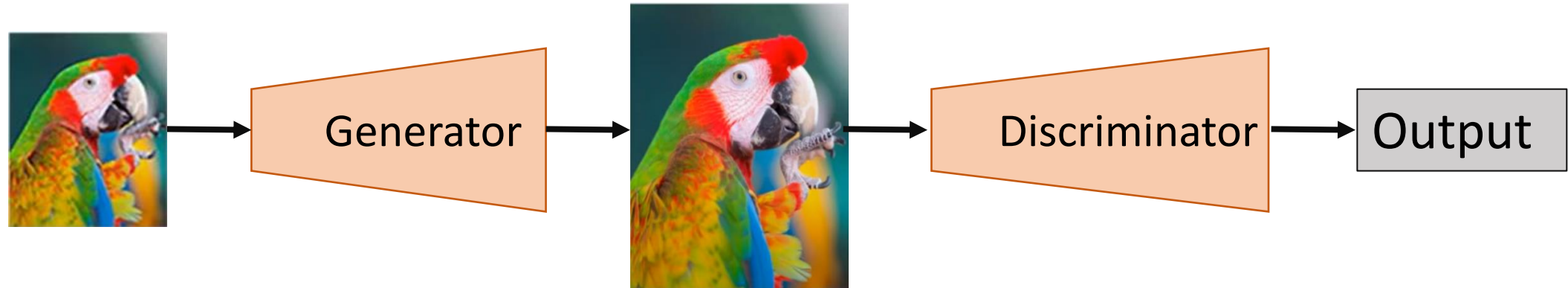
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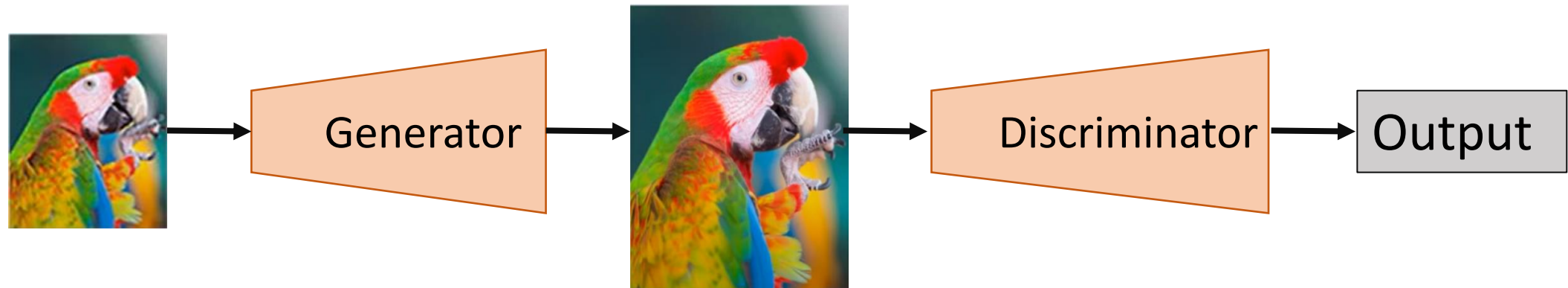
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Super Resolution GAN



- Interpolation techniques for upsampling : bilinear, nearest neighbour, bicubic.
- Super resolution using CNN: Good alternative to using traditional filters.
- Super resolution ResNets: Deeper CNN for better performance.
- SRGANs: Generative models for more realistic super resolution.



Super Resolution GAN

- **Generator:** Low resolution to super resolution image.
- **Discriminator:** Tell super resolution images from original high resolution images.



Perceptual Loss

- **Generator:** Low resolution to super resolution image.
- **Discriminator:** Tell super resolution images from original high resolution images.



Perceptual Loss:

- The perceptual loss is a combination of both adversarial loss and the content loss.

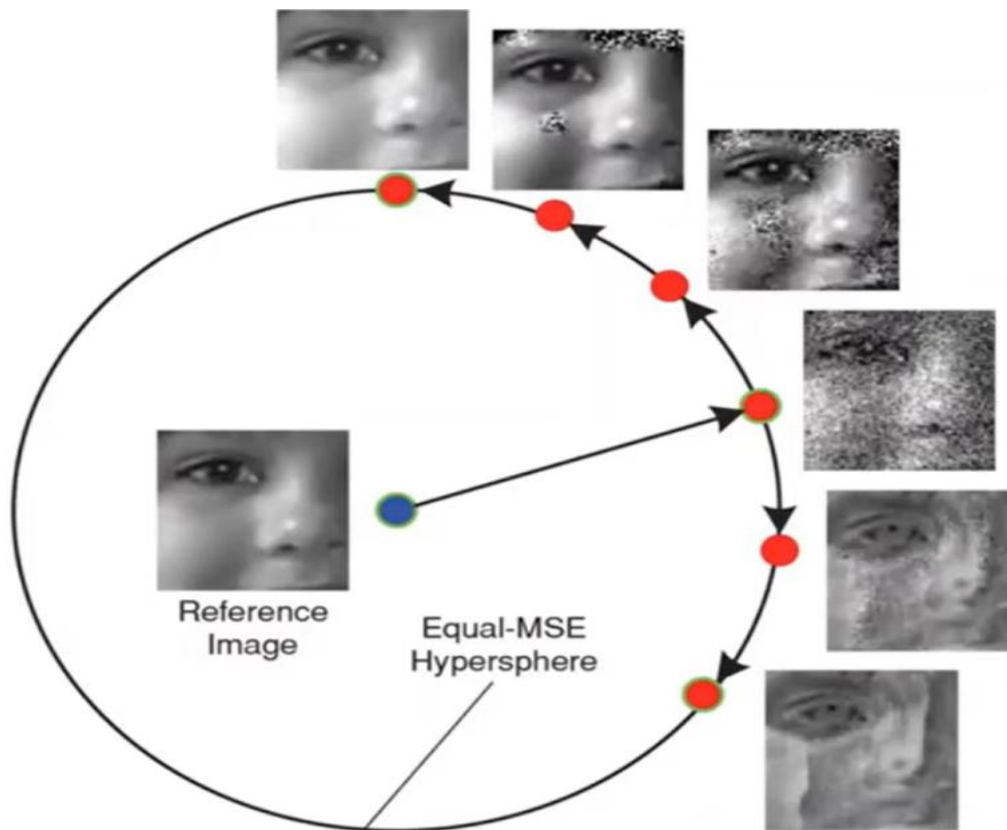
↓
Discriminator loss

↓
Content loss

Ref. <https://www.youtube.com/watch?v=KsNLxBvJBKo>

Content Loss

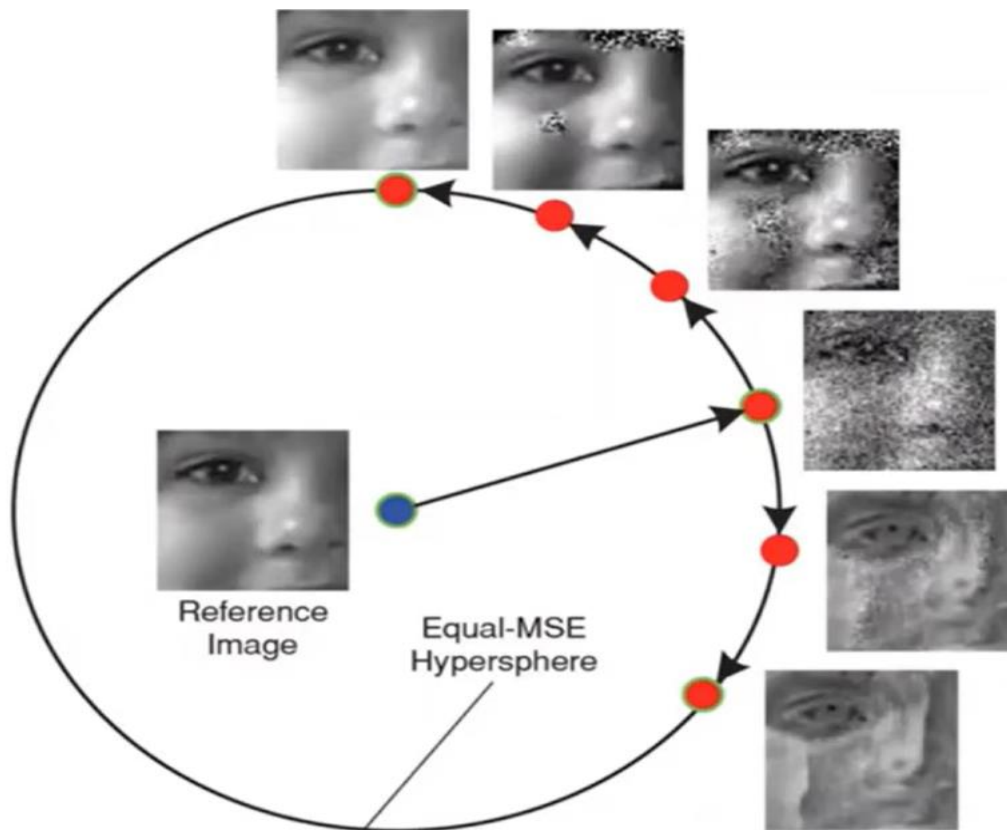
- **Content Loss:** Evaluates the image quality based on its perceptual quality.



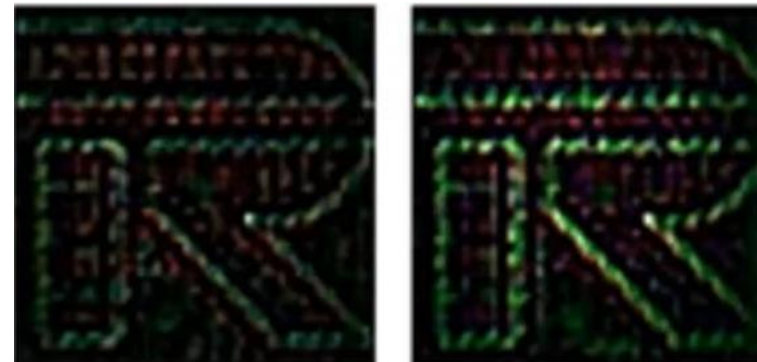
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Content Loss

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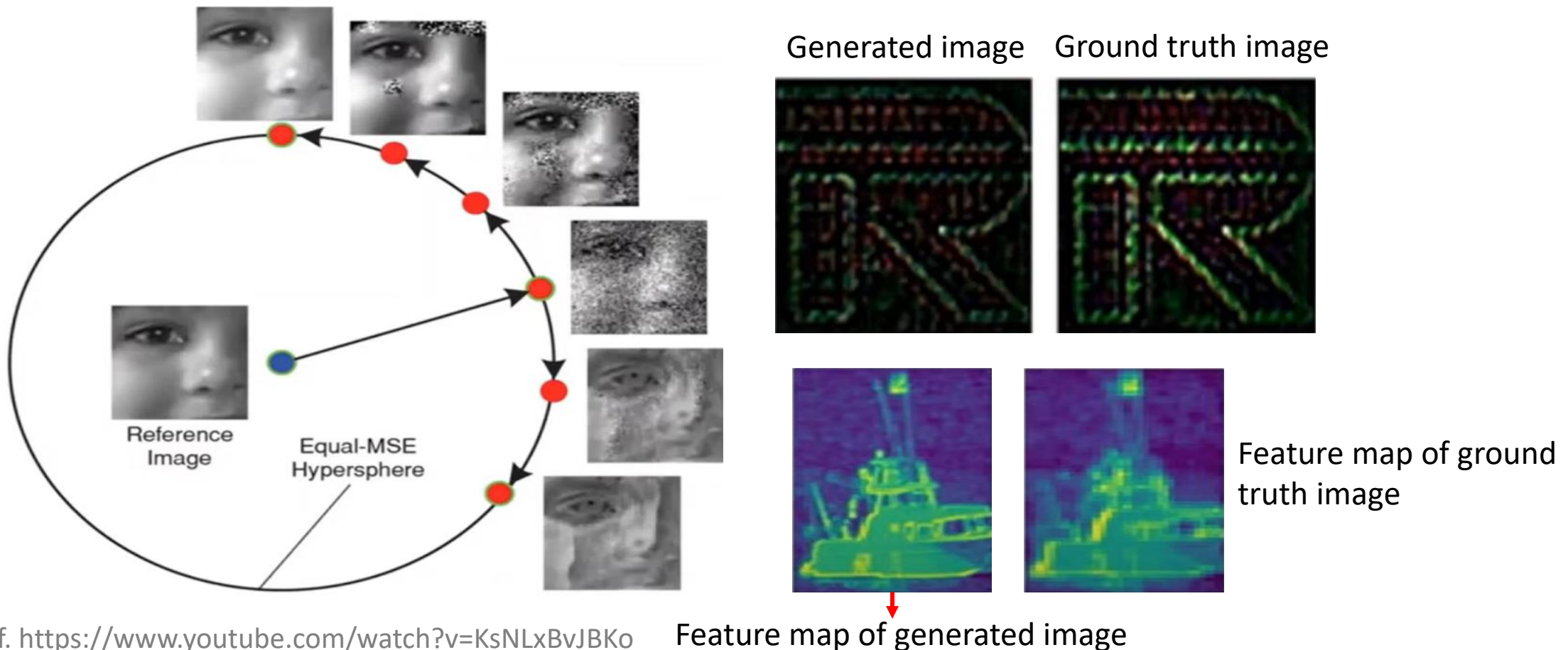
Generated image Ground truth image



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Content Loss

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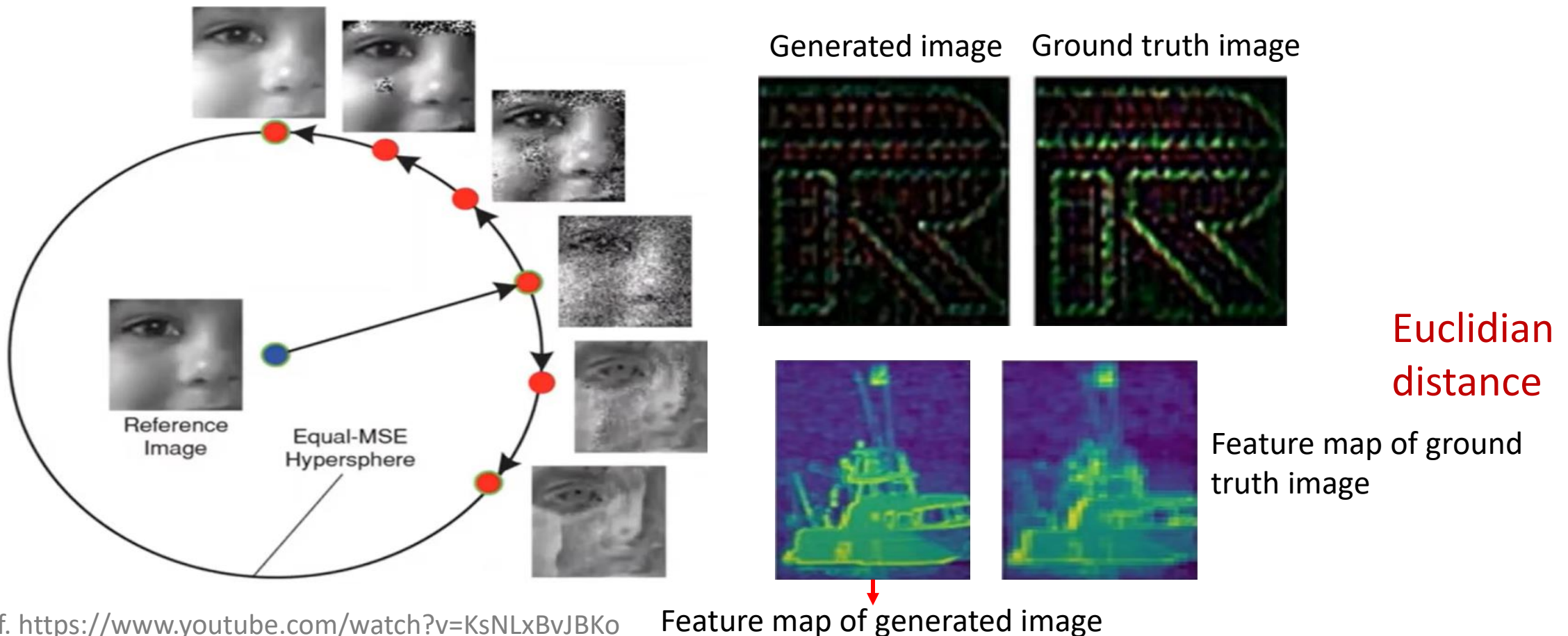


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Feature map of generated image

Content Loss

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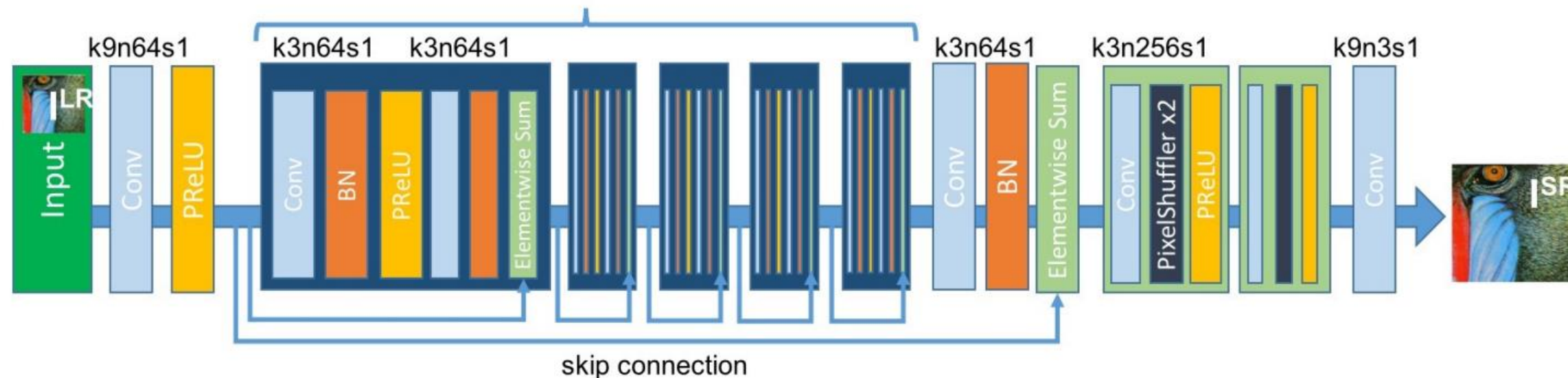


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Feature map of generated image

SRGAN Generator Network

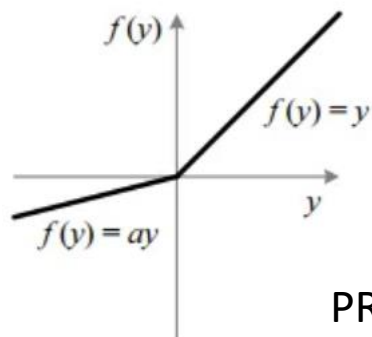
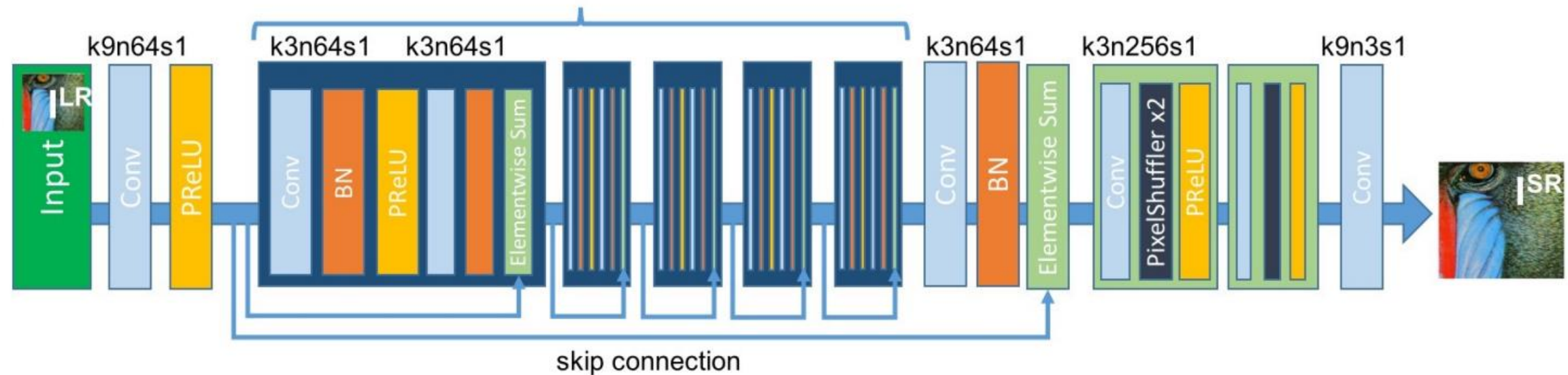
Generator Network



Ref. <https://doi.org/10.48550/arXiv.1609.04802>

SRGAN Generator Network

Generator Network

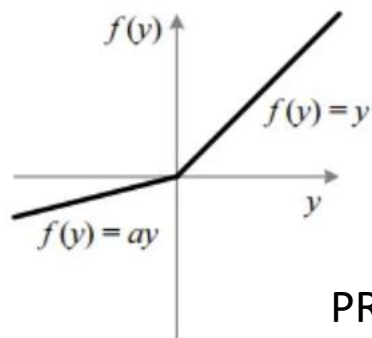
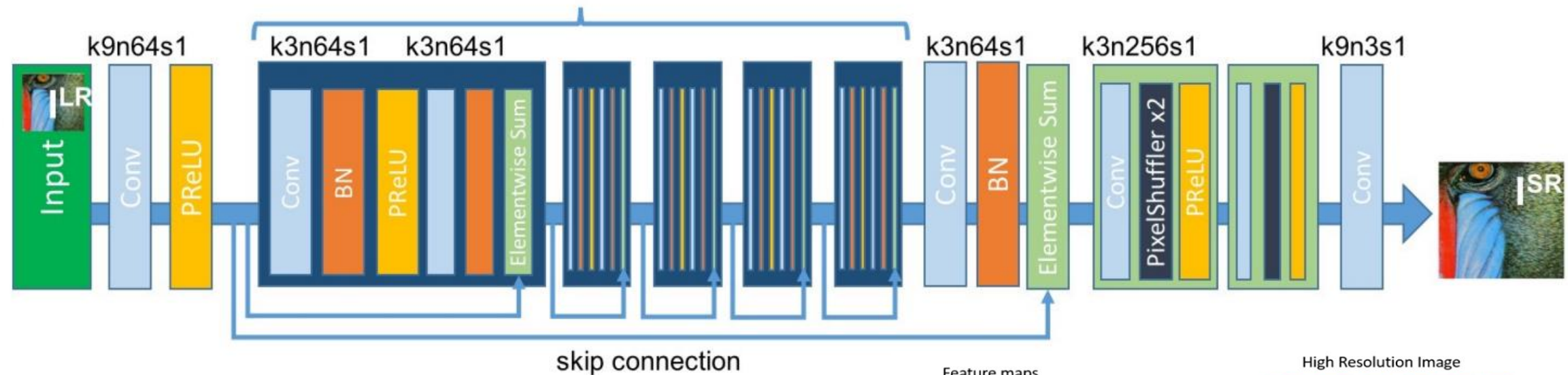


PReLU

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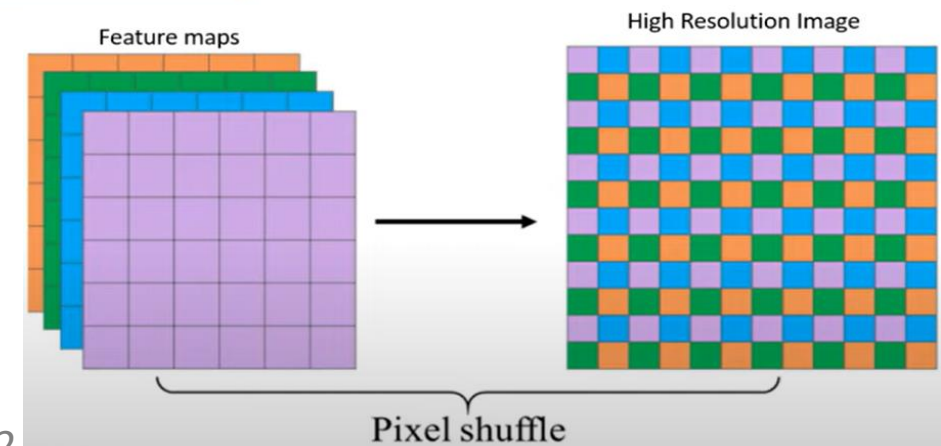
SRGAN Generator Network

Generator Network



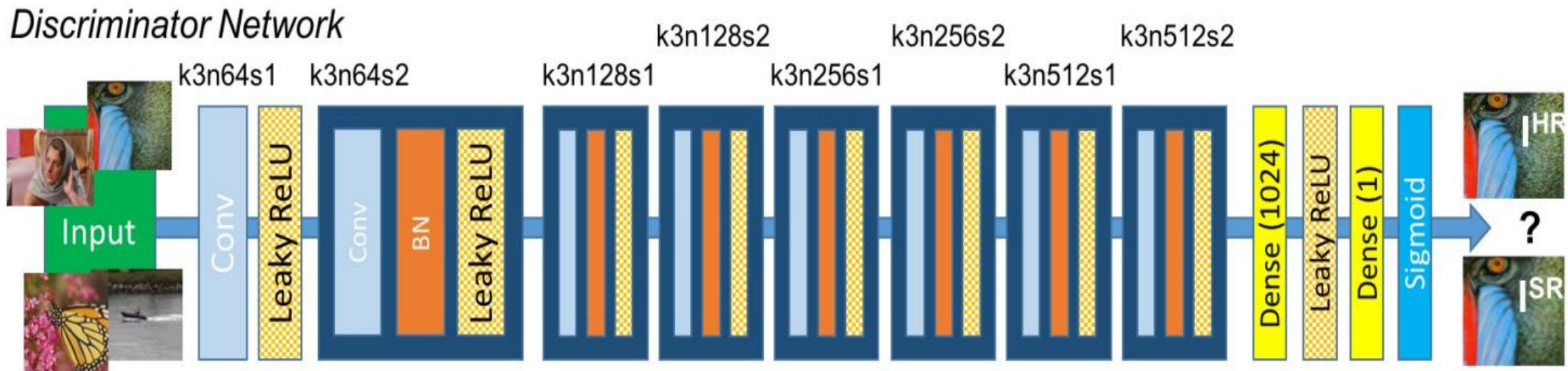
PReLU

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SRGAN Discriminator Network



Ref. <https://doi.org/10.48550/arXiv.1609.04802>



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- **High-quality images:** SRGAN can generate high-quality images with significantly higher resolution and finer details than the low-resolution input images.





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- **High-quality images:** SRGAN can generate high-quality images with significantly higher resolution and finer details than the low-resolution input images.
- **Realistic textures:** SRGAN can produce images with realistic textures, which makes the generated images more natural-looking.
- **Adaptable to different tasks:** SRGAN can be adapted to different tasks such as image restoration, denoising, and deblurring.



Disadvantages of Super Resolution GAN

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- **Limited training data:** SRGAN requires a large amount of high-quality training data to learn to generate high-quality images.
- **Mode collapse:** SRGAN may suffer from mode collapse, which is a common problem in GANs where the generator produces a limited set of outputs, resulting in a lack of diversity in the generated images.



References

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2. Goodfellow, Ian NIPS 2016 Tutorial: Generative Adversarial Networks, NIPS (2016).
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